



Implementation Plan: Dynamic Domain Skins



Goal

Enable VyaparMind to dynamically change its **Visual Identity** (Icons, Colors, CSS) and **Terminology** (e.g., "Table" vs "Bed") based on the configured configuration (Domain).

Core Architecture

1. `domain\_manager.py` (New Module)

This module will serve as the "Style Dictionary". It will export a function `get\_domain\_config()` that returns the assets for the current active domain.

Structure:

```
```python
DOMAINS = {
 "RETAIL": {
 "app_name": "VyaparMind Retail",
 "theme_color": "#E35205", # Sabc Orange/Generic
 "terminology": {
 "inventory": "Inventory",
 "product": "Product",
 "customer": "Customer",
 "table": "Table"
 },
 "icons": {
 "inventory": "■",
 "pos": "■",
 "dashboard": "■"
 },
 },
 "PHARMA": {
 "app_name": "VyaparMind Pharma",
 "theme_color": "#009FDF", # Medical Blue
 "terminology": {
 "inventory": "Pharmacy Stock",
 "product": "Medicine",
 "customer": "Patient",
 "table": "Bed"
 }
 }
}
```

```
},
"icons": {
 "inventory": "■",
 "pos": "■",
 "dashboard": "■"
}
}
```

**Future: HOSPITAL, LOGISTICS etc.**



```
}
...

```

## 2. Global State Management

We will use ``st.session_state['domain']`` to track the active domain. Defaults to 'RETAIL' (or read from ``.env``).

## 3. UI Component Updates (``ui_components.py``)

- **Sidebar:** Instead of hardcoding ``st.page_link("pages/1_Inventory.py", icon="■")``, it will look up ``domain_manager.get_icon('inventory')``.
- **Headers:** The ``ui.render_top_header()`` function will apply the specific ``theme_color`` to the gradient.

## 4. Page Refactoring Strategy

We will use a "Soft Refactor" approach. Instead of changing every string in every file immediately, we will create a helper:

```
`T = domain_manager.get_terminology()`
```

Then in ``1_Inventory.py``:

```
`st.title(f"{T['inventory']}")` instead of `st.title("Inventory")`.
```

## Proposed Changes

### [NEW] ``domain_manager.py``

- Define the dictionaries.
- Implement ``get_active_domain()``, ``get_text(key)``, ``get_icon(key)``.

### [MODIFY] ``ui_components.py``

- Import ``domain_manager``.
- Update ``render_sidebar()`` to use dynamic icons and labels.
- Update ``render_top_header()`` to use dynamic colors.

### [MODIFY] ``app.py``

- Add a visible "Domain Switcher" (for demo purposes) in the login screen or sidebar to easily toggle skins.

## Verification Plan

1. **Manual Test:** Login, toggle domain to "PHARMA".
2. **Check Sidebar:** Verify "Inventory" icon becomes ■.
3. **Check Color:** Verify header gradient shifts to Blue.

4. **Check Text:** Verify "Customers" page is referred to as "Patients" (if implemented).